

Wave Dynamics on the Critical Möbius Strip: Spectral Collapse, Phase Delay, and an Emergent A_2 Root Configuration

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Abstract

We investigate the spectral and dynamical properties of Hodge-harmonic 1-forms on the Schwartz critical Möbius strip, characterised by the length-to-width ratio $\lambda/w = \sqrt{3}$ and the curvature–torsion coupling $\kappa = 2\tau$. We exhibit a sharp dichotomy between the one-dimensional asymptotic approximation and the exact two-dimensional embedding. While the thin-strip limit ($\kappa \rightarrow 0$) yields a purely algebraic spectral weight $G_0 = 8\pi^2\sqrt{3}$, the exact non-uniform Gaussian curvature induces a substantial spectral collapse, with the effective eigenvalue dropping to $\lambda_{\text{eff}} \approx 20.734$. By treating the aspect ratio $r = \lambda/w$ as a free parameter, we find a numerical stability threshold $r^* \approx 2.704$ below which the lowest mode ($N = 1$) becomes unstable; the Schwartz embedding $r = \sqrt{3}$ lies below this threshold, so the $N = 1$ mode does not occur there. We further analyse the exact standing-wave eigenfunction under the non-uniform effective propagation speed $c_{\text{eff}}(s)$ and show that traversal through the geometric folds produces a non-vanishing phase delay $\delta\phi(s)$, breaking node–antinode symmetry of the wave. Finally, by integrating the Born amplitude $\sin^2(k_s s)$ of the $N = 3$ mode against $e^{i\tau s}$ over six equal panels of the orientable double cover, we obtain six complex amplitudes $z_1, \dots, z_6$ with $z_{n+3} = -z_n$ and $\sum z_n = 0$. As an abstract root system in the plane, this configuration is isomorphic to the A_2 root system underlying $\mathfrak{su}(3)$; concretely, the six vectors lie at $\{30^\circ, 90^\circ, \dots, 330^\circ\}$, rotated by 30° relative to the standard simple-root basis. The construction yields the root locations of A_2 but does not, by itself, recover the full Lie-algebraic structure of $\mathfrak{su}(3)$.

1 Introduction

1.1 The Geometry of the Schwartz Möbius Strip

The Möbius strip is the simplest non-orientable surface, and the spectral analysis of differential operators on it provides a natural test case for the interaction between non-orientable topology and Riemannian geometry. We focus on the Schwartz critical embedding [1], the smooth Möbius band of minimal aspect ratio $\lambda/w = \sqrt{3}$. At this embedding, the centre line satisfies the equilibrium constraint $\kappa = 2\tau$ relating the bending curvature to the torsion [2], and the surface carries non-uniform Gaussian curvature $K = -\tau^2$ along the centre line; in particular it is strictly non-developable. The present paper studies the spectrum of the Hodge–de Rham Laplacian on 1-forms over this surface, and the discrete algebraic structure that emerges from its $N = 3$ resonant mode.

1.2 The Spectral Dichotomy: Asymptotic vs. Exact

Previous analyses have often relied on the thin-strip asymptotic limit ($w \rightarrow 0$, $\kappa \rightarrow 0$), in which the wave equation reduces to a constant-coefficient ordinary differential equation. In this idealised one-dimensional limit, the eigenvalues take simple algebraic forms; in particular, one obtains the asymptotic spectral weight $G_0 = 8\pi^2\sqrt{3}$.

We show that this asymptotic approximation differs substantially from the spectrum of the exact two-dimensional embedding. The non-uniform metric induced by $\kappa = 2\tau$ acts as a strong geometric impedance, and the effective eigenvalue collapses to $\lambda_{\text{eff}} \approx 20.734$. Moreover, treating the aspect ratio $r = \lambda/w$ as an independent geometric parameter, we observe numerically that the lowest ($N = 1$) mode has $\lambda_1(r) < 0$ for $r < r^*$ with $r^* \approx 2.704$; the Schwartz value $r = \sqrt{3}$ lies in this regime. The lowest mode is therefore not available at the Schwartz embedding, and we focus on the $N = 3$ mode, whose eigenvalue is verified numerically to be positive.

1.3 Geometric Impedance and Phase Delay

Beyond the static spectrum, the curvature gradient alters the local wave dynamics. The transverse-averaged metric defines an effective propagation speed $c_{\text{eff}}(s)$ along the centre line, with global minima at the geometric folds (regions of maximal curvature). When a standing wave traverses such a slow zone, it acquires a non-zero phase delay $\delta\phi(s)$ relative to the constant-speed reference. We derive an exact Fourier identity (Theorem 1) showing that the phase-delay difference between symmetrically placed anti-nodes carries a non-vanishing algebraic factor $\sin(2n\pi/3)$. As a consequence, the Born density $\sin^2(\varphi(s))$ is asymmetrically distributed across the fold, and the energy density of the resonant wave is displaced from its flat-space baseline.

1.4 Emergent A_2 Root Configuration

The interaction between the non-orientable topology and the $N = 3$ mode produces a discrete algebraic structure. Since the Möbius strip $\mathcal{M}$ is single-sided, we work on its orientable double cover $\mathcal{DT}$ and partition the cover into six equal panels of length $\lambda/3$. The Born panel amplitudes

$$z_n = \int_{(n-1)\lambda/3}^{n\lambda/3} \sin^2(k_s s) e^{i\tau s} ds, \quad n = 1, \dots, 6,$$

satisfy a 60° recurrence $z_{n+1} = e^{i\pi/3} z_n$, the antipodal relation $z_{n+3} = -z_n$, and the closure condition $\sum_{n=1}^6 z_n = 0$. The six vectors lie at $\{30^\circ, 90^\circ, \dots, 330^\circ\}$ in the complex plane and form, as an abstract root system, a copy of A_2 rotated by 30° relative to the standard simple-root presentation. We emphasise that this construction yields only the angular configuration: the Cartan decomposition, simple-root choice, step operators, and Lie bracket relations of $\mathfrak{su}(3)$ are not part of the present analysis.

2 Hodge–de Rham Spectra: Asymptotic vs. Exact Geometry

To analyse wave dynamics on the critical Möbius strip, we study Hodge-harmonic 1-forms. Treating the centre-line $s \in [0, \lambda)$ as the principal propagation coordinate and averaging over the transverse width $t \in [-w/2, w/2]$, we reduce the eigenvalue problem to a 1D problem modulated by the strip’s geometry.

2.1 The Thin-Strip Asymptotic Limit

In the thin-strip limit $w \rightarrow 0$ with $\kappa \rightarrow 0$ (the manifold treated as a twisted 1D loop), the metric components approximate the Euclidean form $g_{ss} \approx 1$. The wave equation simplifies to a constant-coefficient eigenvalue problem:

$$-f''(s) - \tau^2 f(s) = \lambda f(s). \quad (1)$$

For anti-periodic modes accommodating the Möbius boundary condition $f(\lambda) = -f(0)$, the fundamental wave number is $k_0 = \pi/\lambda$. The N -th mode has $k_s = Nk_0$, with eigenvalue

$$\lambda_{\text{ideal}} = (N^2 - 1)k_0^2. \quad (2)$$

Defining the dimensionless spectral weight $G = \lambda_{\text{ideal}} \cdot \lambda^2 / (\text{Area})$ in Schwartz units $\lambda/w = \sqrt{N}$, the $N = 3$ configuration gives the algebraic constant

$$G_0 = (3^2 - 1) \left(\frac{\pi}{\lambda} \right)^2 \cdot \lambda^2 \cdot \sqrt{3} = 8\pi^2 \sqrt{3} \approx 136.757. \quad (3)$$

This value, however, neglects the transverse geometric impedance present in the true spatial embedding, as we now show.

2.2 Exact Curvature Correction and Spectral Collapse

In the exact embedding, $\kappa = 2\tau$, and the longitudinal metric component becomes

$$g_{ss}(s, t) = 1 + 2t\kappa \sin(\tau s) + t^2 [\tau^2 + \kappa^2 \sin^2(\tau s)]. \quad (4)$$

The non-uniform Gaussian curvature $K = -\tau^2$ produces a non-trivial geometric drag on wave propagation. The effective 1D eigenvalue equation becomes

$$-(c_{\text{eff}}(s)f'(s))' - \tau^2 f(s) = \lambda_{\text{eff}} f(s), \quad (5)$$

where $c_{\text{eff}}(s)$ is the effective propagation speed obtained by integrating $g_{ss}^{-1/2}$ over the transverse cross-section.

For the $N = 3$ mode, the resulting eigenvalue is markedly different from the thin-strip prediction. We verify this with two independent numerical methods. Method A uses a Rayleigh–Ritz expansion in N_b anti-periodic trigonometric basis functions; Method B is a second-order finite-difference scheme on N_s grid points covering $[0, \lambda)$.

Method A (Rayleigh–Ritz)	λ_{eff}	Method B (Finite-Difference)	λ_{eff}
$N_b = 4$ basis functions	20.7382	$N_s = 2000$ grid points	20.7351
$N_b = 6$ basis functions	20.7340	$N_s = 4000$ grid points	20.7342
$N_b = 8$ basis functions	20.7339	$N_s = 8000$ grid points	20.7339
$N_b = 10$ basis functions	20.7339	$N_s = 16000$ grid points	20.7339

Table 1: Convergence of the exact effective eigenvalue λ_{eff} under two independent numerical schemes.

Both methods converge to $\lambda_{\text{eff}} = 20.7339 \dots$ at high resolution. The exact spectral weight is approximately $G_{\text{curv}} \approx 108$, far below $G_0 = 136.757$. The exact non-uniform curvature thus acts as a strong geometric impedance, and the algebraic value G_0 derived in the thin-strip limit is not a good approximation to the spectrum of the exact embedding.

2.3 Topological Instability of the $N = 1$ Mode

The most striking consequence of the curvature correction concerns the lowest mode, $N = 1$. In the thin-strip limit, $N = 1$ yields a trivial zero eigenvalue ($\lambda = 0$). Under the exact metric with $\kappa = 2\tau$, however, the eigenvalue is driven negative:

$$\lambda_1 \approx -0.59 < 0. \quad (6)$$

A negative eigenvalue in the Hodge–de Rham spectrum signifies that the $N = 1$ mode is unstable on the Schwartz Möbius strip: a corresponding wave function would not be normalisable as a stationary mode at this geometry.

To delineate this instability, treat the length-to-width ratio $r = \lambda/w$ as a free geometric parameter and track the lowest eigenvalue $\lambda_1(r)$ numerically. As $r \rightarrow \infty$ (the strip becomes long and narrow), $\lambda_1(r) \rightarrow 0^-$ from below, recovering the flat 1D limit. As r decreases (the

strip widens and curvature gradients steepen), $\lambda_1(r)$ becomes increasingly negative. Numerical root-finding on the exact 1D effective equation gives the zero crossing

$$r^* \approx 2.704. \quad (7)$$

For $r < r^*$, the $N = 1$ mode is not available. Since the Schwartz critical embedding has $r = \sqrt{3} \approx 1.732 < r^*$, the $N = 1$ mode is excluded from this geometry. The $N = 3$ mode at $r = \sqrt{3}$ is verified to have a positive eigenvalue, and we use it throughout as the lowest accessible mode on the Schwartz embedding. We do not address whether intermediate odd modes ($N = 2$ is forbidden by the Möbius boundary condition) are stable at all geometries; the focus of the present paper is the $N = 3$ mode at $r = \sqrt{3}$.

3 Geometric Impedance and Phase Delay

In a uniform 1D medium, standing waves have symmetric node distributions. On the exact Möbius strip, the non-uniform $c_{\text{eff}}(s)$ alters the local wave number and introduces a position-dependent phase delay.

3.1 Phase Delay Formulation

The exact standing-wave eigenfunction can be written in WKB-like form

$$f_{\text{exact}}(s) = \sin(\varphi(s)), \quad \varphi(s) = \int_0^s \frac{k_{\text{eff}}}{\sqrt{c_{\text{eff}}(s')}} ds', \quad (8)$$

with $k_{\text{eff}} = \sqrt{\lambda_{\text{eff}} + \tau^2}$. Define the geometric phase delay $\delta\phi(s)$ as the deviation from the constant-speed reference phase $k_s s$:

$$\delta\phi(s) = \varphi(s) - k_s s. \quad (9)$$

Under $\kappa = 2\tau$, the function $c_{\text{eff}}(s)$ has global minima at the geometric folds (e.g., $s = \lambda/2$). When the standing wave traverses such a slow zone, φ accumulates a negative increment relative to $k_s s$, contributing to $\delta\phi(s)$.

3.2 Fourier Kernel and Spatial Drift

The phase-delay difference between symmetrically placed anti-nodes (e.g., before and after the fold) is governed by a Fourier expansion of the inverse speed. We record the relevant elementary identity:

Theorem 1 (Fourier Kernel). *For any integer $n > 0$,*

$$\int_{\lambda/6}^{5\lambda/6} \cos(2n\tau s) ds = \frac{(-1)^n \sin(2n\pi/3)}{n\tau}. \quad (10)$$

Proof. The primitive is $[\sin(2n\tau s)/(2n\tau)]_{\lambda/6}^{5\lambda/6}$. Using $\tau\lambda = \pi$, the boundary evaluation gives

$$\frac{\sin(5n\pi/3) - \sin(n\pi/3)}{2n\tau}.$$

Applying the sum-to-product identity,

$$\sin(5n\pi/3) - \sin(n\pi/3) = 2\cos(n\pi)\sin(2n\pi/3) = 2(-1)^n \sin(2n\pi/3).$$

Dividing by $2n\tau$ completes the proof. □

Since $\sin(2n\pi/3) \in \{0, \pm\sqrt{3}/2\}$ for all $n \in \mathbb{Z}$, every Fourier mode of the phase-delay difference carries a non-vanishing algebraic factor. Consequently, the Born density $\sin^2(\varphi(s))$ is not symmetric under reflection across the fold: the energy density of the $N = 3$ resonant wave is asymmetrically distributed across the geometric fold, with the displacement governed by $\delta\phi(s)$. This is a quantitative consequence of the non-uniform c_{eff} and is independent of any further physical interpretation.

4 Emergent A_2 Root Configuration on the Double Cover

Combining the $N = 3$ mode with the centre-line torsion τ and the orientable double cover yields a discrete algebraic structure in the complex plane. The Möbius manifold $\mathcal{M}$ (with $s \in [0, \lambda)$) covers half the topological cycle; we work on its orientable double cover, the cylinder $\mathcal{DT}$ with $s \in [0, 2\lambda)$.

We define complex panel amplitudes by integrating the Born weight $\sin^2(k_s s)$ against the local phase $e^{i\tau s}$ over six equal panels of length $\lambda/3$:

$$z_n := \int_{(n-1)\lambda/3}^{n\lambda/3} \sin^2(k_s s) e^{i\tau s} ds, \quad n = 1, \dots, 6. \quad (11)$$

Theorem 2 (Panel Recurrence and Closure). *The vectors z_n satisfy*

1. the 60° recurrence $z_{n+1} = e^{i\pi/3} \cdot z_n$;
2. the antipodal relation $z_{n+3} = -z_n$ (for $n = 1, 2, 3$);
3. the closure $\sum_{n=1}^6 z_n = 0$.

Proof. For each panel, change variables $s = (n-1)\lambda/3 + u$. Since $k_s = 3k_0$ and $k_0\lambda = \pi$, the standing wave amplitude $\sin^2(k_s s)$ has spatial period $\lambda/3$, so $\sin^2(k_s s) = \sin^2(k_s u)$. The phase factor transforms as

$$e^{i\tau s} = e^{i\tau((n-1)\lambda/3 + u)} = e^{i(n-1)\tau\lambda/3} \cdot e^{i\tau u}.$$

With $\tau\lambda = \pi$, the prefactor is $e^{i(n-1)\pi/3}$, and we obtain

$$z_n = e^{i(n-1)\pi/3} \int_0^{\lambda/3} \sin^2(k_s u) e^{i\tau u} du = e^{i(n-1)\pi/3} z_1.$$

This proves (i). Then $z_{n+3} = e^{i\pi} z_n = -z_n$, giving (ii); and (iii) follows from $\sum_{n=1}^6 z_n = \sum_{n=1}^3 (z_n + z_{n+3}) = 0$. $\square$

Remark 1 (Numerical evaluation and angular configuration). *Direct evaluation gives z_1 with phase $\arg z_1 = -60^\circ$ (and modulus determined by the $\sin^2$ integral on $[0, \lambda/3]$). Applying the $e^{i(n-1)\pi/3}$ recurrence, the six vectors lie along the directions*

$$\{30^\circ, 90^\circ, 150^\circ, 210^\circ, 270^\circ, 330^\circ\}$$

in the complex plane. As a configuration of six unit-direction vectors closed under negation and arranged at 60° spacing, this is, as an abstract root system in the plane, isomorphic to the A_2 root system underlying $\mathfrak{su}(3)$. The 30° offset relative to the standard presentation $\{0^\circ, 60^\circ, \dots, 300^\circ\}$ is a global rotation that depends on the choice of starting panel $s = 0$ and is not intrinsic.

Remark 2 (What this construction does and does not establish). *The construction above produces the angular locations of the A_2 root system from a geometric integral. We emphasise the following limitations.*

- *Theorem 2 concerns only the angular and closure structure of the six complex amplitudes z_n . It does not exhibit a Cartan subalgebra, a choice of simple roots, step operators E_α , or the Lie bracket relations $[E_\alpha, E_\beta] = N_{\alpha,\beta}E_{\alpha+\beta}$ that define $\mathfrak{su}(3)$ as a Lie algebra. We have established a root configuration, not a representation of $\mathfrak{su}(3)$.*
- *Whether the present construction can be promoted to a full Lie-algebraic structure—for instance, by interpreting the panel amplitudes as matrix elements of operators on a suitable Hilbert space—is left as an open question.*
- *The appearance of an A_2 configuration here is geometric: it arises from the threefold panel partition, the Möbius closure $\tau\lambda = \pi$, and the spatial period $\lambda/3$ of $\sin^2(k_s s)$. We do not claim a connection to any physical $\mathfrak{su}(3)$ symmetry (e.g. flavour or colour).*

5 Conclusion

We have analysed Hodge-harmonic 1-forms on the Schwartz critical Möbius strip ($\kappa = 2\tau$) and exhibited three independent geometric phenomena. First, the exact non-uniform curvature induces a substantial spectral collapse: the effective eigenvalue $\lambda_{\text{eff}} \approx 20.734$ at $r = \sqrt{3}$ differs markedly from the thin-strip value, and the lowest mode $N = 1$ has $\lambda_1(r) < 0$ for $r < r^* \approx 2.704$. The Schwartz embedding lies in this regime, so the $N = 1$ mode is not available there; we work with the $N = 3$ mode, which has positive eigenvalue. Second, the non-uniform effective propagation speed $c_{\text{eff}}(s)$ produces a non-zero phase delay $\delta\phi(s)$ across the geometric folds, breaking node–antinode symmetry and asymmetrically distributing the Born density across the fold. Third, the Born panel amplitudes of the $N = 3$ mode on the orientable double cover form, as an abstract root system in the plane, a copy of the A_2 root system (rotated 30° from standard presentation), giving a geometric realisation of a rank-2 root configuration on a non-orientable surface. Promoting this configuration to a full Lie-algebraic structure—and clarifying whether the threshold r^* admits an analytic expression—are natural questions left open here.

References

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